# SOLPS-ITER v.3.0.7 Release Notes

as of 22 June 2020

These Release Notes detail changes between SOLPS-ITER versions 3.0.6 and 3.0.7 and give information about how to transition an older SOLPS-ITER run to be continued with the 3.0.7 code version. Users are urged to read this document carefully and thoroughly before proceeding with use of SOLPS-ITER 3.0.7, at least up to and including section 6 which provides instructions on how to transition a case to v.3.0.7 from v.3.0.6. Users should also consult the updated version of the SOLPS-ITER manual that accompanies this distribution and other documents posted to the SOLPS-ITER web site. For full details of all updates, please consult the GIT repository logs. In addition to the items listed below, the new code version also contains lots of bug fixes, new documentation, and output clarification/prettifying improvements, as reported/requested by users and developers over the last several months. Some of the changes that have occurred on the Eirene side are documented in the updated Eirene manual, which can be found at [www.eirene.de](www.eirene.de) .

A Slack discussion group has been created, which you can join at:
[https://join.slack.com/t/solps/shared_invite/zt-ctpoedhs-KZYjYeBDDH7I3pH1M6eGcg](https://join.slack.com/t/solps/shared_invite/zt-ctpoedhs-KZYjYeBDDH7I3pH1M6eGcg) .

If you are transitioning from SOLPS-ITER v.3.0.5 or earlier to v.3.0.7, then you should also read and consult the v.3.0.6 Release Notes.

SOLPS-ITER is now available to users in Canada, as the Canadian government has signed an MoU with the ITER Organization. Usage of the code in the United Kingdom and its dependencies past 31 December 2020 is subject to the user's host institution having signed a scientific collaboration agreement with the ITER Organization.

## 1. How to obtain SOLPS-ITER v.3.0.7:

In order to update your copy of the SOLPS-ITER code suite from version 3.0.6 to 3.0.7, use the "*solps-iter_update*" script. If coming from version 3.0.5 or older, also run the "*switch_to_master*" script. As of version 3.0.6, the SOLPS-ITER distribution does, by default, point to the master branch of the SOLPS-ITER GIT repository and its submodules. If you wish to point instead to the develop branch, then use the "*solps-iter_update_develop*" script after the initial use of "*solps-iter_update*". As usual, before making any such update, please make sure that any local changes you might have are committed to local GIT branches or stashed, so as to avoid them being overwritten.

Users that also need the SOLPS4-5 converter program *b2sxdr* should update their copy of SOLPS4.3 as well.

For subsequent updates, users should use "*solps-iter_update*" or "*solps-iter_update_develop*", depending on which branch they are using, and only once per update.

SOLPS-ITER v.3.0.7 combines, among others, work that has proceeded in the *feature/radiation_tallies*, *feature/OpenMP*, *feature/FZJ_synchronization* and *feature/fix_recomb+redef_gmtry* branches of the repository.

Because the update to v.3.0.7 involves new *use* modules (in particular, if you want IMAS support, you should point to IMAS/3.20.0 at least), and some number of added/renamed/removed source code files, it is likely that the compilations launched by the update scripts will fail the first time. If this is the case, start afresh in a new window, load the SOLPS-ITER environment as usual by means of the *setup.csh* file, and try again to compile. If it fails once more, then do a "*gmake clean*" first and compile again. Recall that you can use *"gmake clean_XXX"* to remove all objects associated with any *XXX* build, for example *"gmake clean_solps_mpi"*.

The configuration files have also been updated for cases where the MPI compiler `mpif90`, based on `gfortran`, is different from the default compiler, either `ifort64` or `pgf90`. As much as possible, we then correct the compilation flags and pre-processor definitions to match the local `gfortran` specifications. Please check these settings for your environment and send us back any changes you want made or missing flags.

2. **<u>Changes to the physics model:</u>**

Please be mindful that the changes to the default physics model described in this section will most likely cause some evolution of existing 3.0.6 solutions. How much will depend on the specifics of the cases at hand. The benchmarks and sample cases available on the SOLPS-ITER website all now have been recomputed using the 3.0.7 physics model.

a. **<u>Numerical constants:</u>**

The physics constants used by the code have been updated to use the CODATA 2018 standard. The ionization energy of Helium in Eirene and AMDS files has been correctly slightly from 24.55 to 24.5874 eV to be consistent with B2.5 values and to improve the calculation of the Eirene radiation tallies.

b. **<u>Viscosity currents:</u>**

The various viscosity currents each have their own controlling multiplier. By default, these multipliers have now all been set to 1.0, but remain subservient to the setting of the "*fac_vis*" drift multiplier, to make it easier for users to turn them all on together in a self-consistent fashion with a single switch setting.

A new variable "*add_te_corr_to_po(0:NREG)*" in *b2.numerics.parameters* allows to not apply the correction to the electric potential due to an electron temperature change until after the perpendicular viscosity current has been recomputed, to avoid numerical instabilities. By default, the switch is set to *.true.*, and the user should consider changing it to *.false.* if the numerical instability, manifested by electric potential and electron temperature oscillations just inside the separatrix in the core region, appears.

The ion-neutral friction conductivity contribution is now only taken into account if drifts are activated, which include the ion-neutral friction current, to ensure consistency of the physics model.

**c. <u>Contributions to kinetic energy from drift velocities:</u>**

The kinetic energy computed by the code only contains the contribution from the parallel component of the velocity. Contributions from the drift velocities are not included, but they are included, by default, to the background fluid velocity passed to Eirene. These contributions can be removed by setting "*b2tfrn_xvecrb*" and "*b2tfrn_xwdia*" to 0.0, for the **E×B** and diamagnetic velocities, respectively.

**d. <u>Inclusion of toroidicity effects in the parallel momentum equation:</u>**

The default value of the "*b2mndr_hz*" switch has been changed from 0.0 to 1.0. This means that toroidicity effects in the parallel momentum equation are now included by default.

**e. <u>Atomic physics sources and sinks:</u>**

The switch "*b2stel_phm0*" has changed meaning to be a multiplier of sources due to atomic physics (new default is 1.0). Its role is no longer restricted to only three-body recombination potential energy losses.

When using locally modified ADAS data files, it is no longer necessary to recopy all the files required by the run (including the unmodified ones) to the *data.local/adas* directory. Only the files which are modified need to be present there, as the code will now look in the standard *modules/adas* location for the original files when locally modified copies are not found. The recommended ADAS datasets for radiation have been updated to "year 42" for Ar, Fe, Kr, W, and Xe, the ionization and recombination data for W changed to "year 50", and the charge-exchange data for N to "year 96". The default reactions for Carbon and Nitrogen volumetric recombination in **Uinp** have also been changed from `AMJUEL 2.3.6A0` and `2.3.7A0` respectively to the ADAS `acd` and `prb` rates. The ionization rate for $Ne^{+5}$ in the ADAS'96 files `scd96` and `scd96r` was also corrected after an error was identified in the underlying data. Moreover, the `b2frates` file header, if using

ADAS, will now include a GIT hash to point to the exact version of the data used to build it. This header information will also be written as part of the `radiation` IDS identifiers.

It is now possible to limit the effect of using the *"b2ardr_fix_recomb"* switch, along with ADAS rates in `b2ar.dat`, by means of *"b2ardr_zmax_recomb"*, to only the isonuclear sequences with a nuclear charge number up to the *zmax_recomb* value, inclusively. The *"fix_recomb"* treatment, still experimental, aims to provide a more accurate calculation of the electron energy gained or lost from recombination events by considering recombination and re-excitation from intermediate excited states. The default values for the *"fix_recomb"* and *"zmax_recomb"* switches are `0` and `1`, respectively, so this treatment is not activated unless requested explicitly.

The code now supports use of the AMNS library, in either its IMAS or ITM versions. When using the former, the `-DAMNS` pre-processor definition but be added in the configuration files. If the latter, then the definition to add is `-DGOT_AMNSPROTO`. These have been added by default for the ITER, ITM, IPP, IPP-HGW, and MARCONI environments. In both cases, please also make sure that you are linking to the AMNS library (preferably version 1.3.2 or higher) and including its definition modules during compilation. The library can be used to build the `b2frates` file by setting the *'flag'* variable in `b2ar.dat` to *'amns'*.

### f. Momentum sources:

The parallel momentum source term "*smban*", which correspond to the **j×B** force due to the anomalous current, is no longer always active but rather turns on only if both the **E×B** and diamagnetic terms are turned on.

### g. Plasma sound speed definition and ion and total pressure distinctions:

The definition of the plasma sound speed, which is computed as the square root of the total pressure divided by the mass density, has been modified so that only ionized species are included in the pressure and mass density sums. This has the largest impact on sheath boundary conditions for runs with fluid neutrals where the neutral densities are significant compared to the ion densities (such as cases where target detachment occurs). Several of the output files that used to provide the total pressure "*pr*" (which includes contributions from fluid neutrals) now provide instead the plasma pressure "*pz*" (including only contributions from charged species), or both.

### h. Backward compatibility to SOLPS5.0:

Extensive checks were made to make sure that, with the right set of switches (found in *runs/examples/b2mn.dat.5.0*), previous results from SOLPS5.0 runs can be recovered exactly. New switches introduced for that goal are "*b2tral_cvsa_style*" and "*b2tqin_csigin_style*", which both need to be set to 0 to recover SOLPS5.0 behaviour, while their default SOLPS-ITER value is 1.

### i. Fluid neutral flux limits:

The default settings for the switches "*b2tlc0_alpha*", "*b2tlc0_gamma*", "*b2tlh0_alpha*", and "*b2tlh0_gamma*" have all been changed to "1.0". This now enforces a flux limit for the convective and conductive transport due to the fluid neutral species.

### j. Total energy fluxes:

The computation of energy fluxes in the `b2xpen` routine has been reviewed and corrected, to avoid double-counting contributions from anomalous fluxes. The energy fluxes `fht`, `fhj`, `fhm`, and `fhp` are now also saved in `b2fplasma` so they can be retrieved for post-processing. They contain, respectively, the total energy flux, the electrostatic energy flux, the kinetic energy flux (including viscous energy terms), and the ionization potential energy flux. The code has also been simplified so that it is these fluxes that are used in diagnostic files and to compute things such as wall and target heat fluxes.

## 3. Changes to the input files:

### a. Enforcing constant poloidal and toroidal magnetic flux:

The new default behaviour of **b2ag** (using the "*b2agdr_redef_gmtry*" switch, set to "*1*" by default) is now to enforce a constant poloidal magnetic flux for each flux tube, as well as make sure that the product of the toroidal field and the major radius stays constant, no matter whether one uses magnetic field values at cell centres, cell faces, or cell corners. The effect of this switch can be seen by the "sanity checks" at the end of **b2ag** indicating that *hx*, *hy*, and *pbs* have changed by something of order 1% or so. Please also remove any mention of the "*b2mndr_redef_pbs*" switch from your *b2mn.dat* files, if any, as this now is superseded by "*b2agdr_redef_gmtry*" in *b2ag.dat* instead and the code will stop if the former switch is found. This treatment has also been corrected for PFR flux tubes, where a bad initial choice in the reference cell led, in some cases, to an unphysically large current in the flux tubes immediately below the separatrix in the PFR. Users seeing such features are therefore encouraged to rebuild their *b2fgmtry* files by re-running **b2ag** before proceeding with their runs. Do not forget, of course, then to also update the *b2fstati*, *b2fpardf*, and *b2frates* files, and to check, by means of the dry-run command "*b2run –n*", that you will not be restarting from a flat state.

To take full advantage, in drift runs, of the numerical magnetic flux conservation properties afforded by building your geometry file with *"b2agdr_redef_gmtry*" set to *"1"*, one should also change the settings of *"b2tfnb_drift_style"*, *"b2tfnb_PSch_style*", and *"b2tiner_fch_inert_style"* from *"1"* to *"2"*, which will provide for a more accurate computation of the drift fluxes. These are the new default values. To recover the old default behaviour in versions 3.0.6 or older, *"b2tiner_fch_inert_style"* needs to be set to *"0"*. These new defaults will cause a small change in the solution, even in cases without drifts, because of the more accurate geometrical prefactors now used when translating currents into particle fluxes.

An additional correction is now applied to the "*qc*" calculation (the cosine between a cell's left face and the magnetic field) to obtain a more accurate value. This is achieved by means of the "*b2agmt_qc_style*" switch, now set to "*1*" by default. Another option, but not quite as accurate, can be obtained with use of "*b2agdr_redef_qc*" set to "*1*", but this is not the recommended choice.

**b. Default location:**

For most input files, they will be searched in order, in the *b2mn.exe.dir* directory, in the directory from which the run is launched, and in putative *baserun* directories at the same level and one level above of the launch directory.

Also, when using the *open_file* routine, the file will be looked for, in addition to any local and explicit paths given in the argument list, at the location corresponding to having a *$SOLPSTOP/* stem added to the filename.

When using bundled ADAS data, although the location of that data must be provided in the *ratadas.filelist* file, any data not found at that location will then be searched for in the standard *$SOLPSTOP/modules/adas* location. This applies to **B2.5**, **Eirene**, and **Uinp**.

**c. Eirene:**

A safety was added to Eirene so that, if the input file indicates that the *fort.15* file is to be read, but no such file is found or it is empty, the code does not fail, but rather writes out a warning message and proceeds with an empty census array. Similarly, there is a safety when reading the *fort.14* file for the code to be able to continue if the file is found empty, in which case the *NRECOM* and *XMCT* array are filled with default values.

An automatic test is now performed, after the first Eirene call in a coupled run, that the definitions of the recycling strata in the Eirene input file and the *b2.neutrals.parameters* file match, in terms of both location and species range. It will be enforced that the recycling strata must be listed in the same order in both files, and assumed that they are listed first in *b2.neutrals.parameters*.

The XDR formatting of the *fort.XX* Eirene I/O files is no longer supported. The content of the *fort.13* file has also changed, as the Eirene plasma background input tallies have been reconfigured. This may mean that, in order to restart a case from an older version, removal of the *fort.13* file may be necessary in the first instance. The *fort.15* file, which contains the census stratum, is now rewritten on every Eirene call for time-dependent runs, not just the first one.

The TRCDBG… debug switches from the Eirene input file are now deprecated. The logical in position 16 of the first block 11 line now stands for TRCHKTIM, which provides timing diagnostics from the main Eirene routines. Moreover, the NLTRIMESH logical switch was moved from block 1 to block 14, to avoid clashes with other Eirene developments and put it in the more appropriate section relevant only to SOLPS-ITER coupling data. Both of these changes are backward-compatible, meaning that old input files will still be understood by the code, but both **Uinp** and **b2yt** will now produce input files with the new conventions.

Eirene can now read ADAS data files which include charge state bundling, by adding, on the second line of the reaction card, after the name of the element and the charge state (or bundle number), a string of character that gives the extension of the name of the *adf11* file to be read. For instance, if one wishes to obtain the data stored in, say the `scd42_w_02_6.dat` file, the string to add is `02_6`. The string has to be added starting on the 13th character on the line being read. In addition, if a case is being built, using **Uinp**, with a *ratadas.filelist* file present that calls for bundled charge states, the Eirene input file will include the correct bundling scheme information for the reaction rates. Similarly, when converting a case using **b2yt** whereby the bundling scheme and/or the ADAS data year are being changed, the bundling scheme information will be updated in the Eirene reaction list.

### d. <u>TRIM data files:</u>

The default **Uinp** behaviour has been changed so that the default TRIM file being used for reflection coefficients in Eirene is no longer the large *trim.dat* file, but rather the individual *A_on_B* files that are listed in block 6 of the Eirene input file. These files are found in the *Database/Surfacedata/TRIM* directory of the Eirene submodule, and the code will search for them there by default.

Users should also be aware of the *A_on_B_X.XX* files, placed in the same location. The *X.XX* numbers correspond to the binding energy of the projectile atom on the surface material (in eV). By default, TRIM used a value of 1.00 eV unless otherwise specified in the file. Although no firm recommendation as to which set of TRIM files should be used can be made at this time, users are encouraged to investigate, for their particular runs, the impact of changing the reflection rates by using the various TRIM files available. The effect is expected to be more strongly felt at low incident energies where the TRIM Binary-Collision-Approach approximation breaks down and the binding energy becomes significant compared to the projectile's incident energy.

### e. Boundary conditions and feedback schemes:

Checks have been added to some of the density boundary conditions, when they are not compatible with use of "*eirene_ionising_core*", to make sure that the switch is not activated. When the switch is activated and the boundary condition calls for a poloidal variation of the density or the radial flux crossing the boundary, that same poloidal variation is applied to the return flux of ionized neutrals.

Two new density feedback schemes (*NA_FEEDBACK_CHOICE = 8* and *9*) are now available in *b2.feedback_control.parameters*, with which to apply a feedback on the relative average concentration of an isonuclear sequence averaged over a rectangular region of cells in the computational domain (*8*) or on the inner midplane electron density (*9*), where the inner midplane location is located at poloidal index *jxi*.

### f. Obsolete switches and commands:

The following input switches from *b2mn.dat* have been made obsolete:

- "*b2tvspa_vis_par*" has been superseded by "*b2tfhe_vis_par*".
- *"b2mndr_device"* is now superseded by *"b2mndr_database"*.
- To enforce compatibility of the treatment of the atomic physics rates, the "*b2stel_fix_recomb_energy*" switch has been deprecated and we use the value of the "*b2ardr_fix_recomb*" switch read in from *b2frates* instead.

The NMACH switch in the Eirene input file is now obsolete, as well as the STATIS_COP and STATIS_BGK related quantities.

The "*srun*" command (alias to "*cd $SOLPSTOP/runs/*") has been replaced by "*sbr*" to avoid conflicts with the SLURM command with the same name.

### g. New speed-up schemes:

A new speed-up scheme is available, using the "*b2stbr_sna_corr*" switch (and, accessorily, the "*b2stbr_taumax*" and "*b2stbr_c_pumpavr*" switches), that is useful to quickly move the solution, when a change in gas throughput (through changes in puffing or pumping rates, for instance) is occurring, towards its new particle content equilibrium. Full details are given in the paper by E. Kaveeva *et al.*, Nucl. Fusion **58**

(2018) 126018, which you can find in the "SOLPS-ITER Papers" folder of the SOLPS-ITER SharePoint website.

Moreover, it is possible to damp out the evolution of poloidal variations on closed field lines by means of the "*corr_core_dt*" switch, which allows for the time evolution of fluxsurface-averaged temperatures to proceed normally while slowing their poloidal variation by the value of switch. This permits for a faster way to reach rough equilibrium of the core quantities, particularly for drift cases, when a change in case parameters requires a new overall core plasma solution. Use is similar to the "*corr_core_dn(is)*" array. This latter array has been modified so that the elements related to the fluid neutrals have no effect. The *"corr_core_d[nt]"* switches are part of the namelist given in the *b2.numerics.parameters* input file. One can also limit the depth extent in the core where this damping is applied by means of the "*nrings_for_no_speedup_averaging*" switch, which indicated the number of the outermost poloidal flux rings inside the separatrix where the damping is not to be applied, so as to keep unchanged the evolution of the plasma near the X-point, for instance.

### h. <u>Case build-up:</u>

When building up a case with the "*carre*" and "*triang*" scripts, it is now checked whether the **Carre** grid has been changed to make the default next step then to rebuild the *b2fgmtry* and *fort.30* file before proceeding to the triangulation step, and the scripts will also not continue by default if the **Uinp** programs fails and does not produce all the expected preparatory files. Additional safeguards have also been included in **Uinp** to ensure that the case being prepared is compatible with the dimensions declared in the *DIMENSIONS.F* file with which the SOLPS-ITER code suite has been compiled. The scripts will also look for the *SOLPSTOP* file, as the B2.5-Eirene executables do, to know from which tree, if not the default, their executables should be running. In case the *dg.str* file and associated links are not present, the "*carre*" script asks for the user to provide the directory where it can found before the gridding operations can proceed.

The **Carre** translation program **tradui** will now add, at the end of the **.geo* and **.sno* grid files, information about the poloidal and toroidal flux functions, noted "*fpsi*" and "*ffbz*" in **B2.5**, for use in the code and plotting in **b2plot**. These functions will be rescaled in **b2ag** if the *b2agfs_Bt_rescale*, *b2agfs_pit_rescale*, *b2agfs_Bt_adjust*, *b2agfs_xoffset* or *b2agfs_yoffset* switches are used, in a consistent manner with the magnetic field component rescalings.

A new equilibrium conversion utility program, *fiesta2dg*, has been provided by David Coster for reading equilibrium files provides by the **FIESTA** code.

4. **New features:**

   a. **OpenMP parallelization of B2.5:**

The B2.5 plasma solver has been parallelized by Tamás Feher using OpenMP. To compile the OpenMP parallelized code versions, add *"_openmp"* to the Makefile target, for example: *gmake all_openmp*. The debug option still goes at the end ("*gmake all_openmp_debug*"). One can also compile with both MPI and OpenMP parallelizations activated together, with the *"_mpi_openmp"* or *"_openmp_mpi"* extensions to the Makefile targets.

To direct SOLPS to use the OpenMP compiled executables, use the "*set_openmp*" script. The "*unset_openmp*" script undoes that command.

When submitting a job, specify the number of OpenMP threads to be used by means of the *"-t #"* option, where "*#*" is the requested number of threads. This option is available for both "*b2run*" interactive job launches and "xxx*submit*" job submission scripts.

It is also possible to select on which individual routines the OpenMP parallelization is performed by means of the *–DNO_OPENMP_XXXX* flags in the B2.5 Makefile when compiling the code. The current set of such flags in the distributed version of the Makefile corresponds to the loops that have been successfully tested. Loops that are currently disabled should be treated with care and for testing/debugging/fixing purposes only.

   b. **Eirene MPI parallelization:**

A new NPRLL switch now replaces the deprecated NMACH switch, located as the very first integer in block 1 of the Eirene input file. It specifies the MPI parallelization strategy to be used. If set to `-1`, then the MPI parallelization strategy is set from the `INFORMATION_FOR_MPI` block at the end of the input file as before. This will be the new **Uinp** and **b2yt** defaults to keep the old behaviour. If set to `0`, then the new "embarrassingly parallel" scheme is used. This scheme is similar to the APCAS (*All Processors Compute All Strata*) scheme, but assigns the full number of particle trajectories to each processor instead of dividing them up. If set to `1`, `2`, or `3`, then the selected MPI parallelization strategy will be "ORIGINAL", "APCAS", or "BALANCED", respectively. The `INFORMATION_FOR_MPI` block is now optional and is used to overwrite the `NPRLL` setting.

The default MPI parallelization strategy for Eirene is now enforced to be, if compiled with MPI version 3 or above, the "BALANCED" strategy, and, if compiled with an older MPI version, the "ORIGINAL" strategy. Both strategies share the workload by assigning full strata to individual processors when possible, or split the strata onto multiple processors. The “ORIGINAL” strategy will do the splits evenly, while the “BALANCED” strategy will use information from the previous Eirene call, if available, to take into account the actual average CPU time spent per trajectory for each stratum when deciding on the workload distribution.

### c. Support for non-lower-single-null (LSN) configurations:

The various case build-up code components have been updated to provide support for limiter configuration cases. The procedure for setting them up is can be found in Section J.6.6 of the SOLPS-ITER manual.

The code has also been corrected to handle linear cases where the toroidal field is zero. Linear cases with plasma entry on one side and a target on the other should have the target located to the right of the domain for best results. A new paper (M. Sala *et al.*, PPCF **62** (2020) 055005) has been published that details how the code set-up changes for linear cases, and is available on the SOLPS-ITER SharePoint website.

When running a double-null case, the code now requires a *b2.boundary.parameters* file to be present so that information about the upper targets will be certain to be provided.

For upper-single-null configurations (USN), the default definitions of the inner and outer midplane locations (*jxi* and *jxa*) have been modified to match their geometrical locations, as the order of the targets is inverted compared to the standard LSN configuration. Recall that for an LSN (USN) case, the “Western”, or “left”, edge of the grid corresponds to the inner (outer) divertor target while the “Eastern”, or “right”, edge of the grid is the outer (inner) divertor target. The tallies labels have been modified to be unambiguous, and the code will now indicate if it detects a USN topology. This also applies to limiter cases where the plasma tangency point is on the top side. Moreover, the output files from the “`wlld`” command from **b2plot** and from *b2wdat*, as well as the related IDS quantities, will now have labels that correctly reflect their geometrical location. The default values of *jxi* and *jxa* have also been slightly modified for linear and stellarator geometries, see the documentation for details. When converting between a USN and an LSN case, **b2yt** will now mirror the Western and Eastern divertor regions so that inner (outer) divertor data is still mapped to the inner (outer) divertor data of the new topology.

### d. New electric potential equation solution method

The solution method for the electric potential equation has been modified to only require a single pass, so that the older scheme of internal potential iterations is no longer needed. This increases the code’s robustness and provides for some limited code speed-up. The plasma current is computed simply as the sum of the currents coming from separate individual routines, one for each physical mechanism. When using the 5.2 physics model, the electric current is now set in the new routine *b2tfch_.F*.

**e. Support for non-hydrogenic plasmas**

The code has also been modified to be able to handle cases where the plasma does not contain any hydrogenic species. However, several of the physics model terms only apply to plasmas with a dominant hydrogenic component, thus the choice of physics options will be more limited.

**f. IMAS support:**

The current recommended IMAS version to use with SOLPS-ITER is IMAS/3.28.0-4.7.2 and GGD version 1.9.1. The configuration files have been modified to detect the IMAS, GGD and UAL versions present and the relevant code is bracketed with compiler `define` statements to make compilation compliant with the locally available IMAS implementation. Efforts have been made to make the code backward-compatible to older versions of IMAS, both in the source code itself and in the compiler configuration files. If you encounter any problems with the latter, please report the issue to the IO development team so we can provide a fix that will fit with the overall logic of these files. If not already provided, please define the environment variable `GGD_VERSION` in your *$SOLPSTOP/SETUP/setup.csh.$HOST_NAME.$COMPILER* file, for example with "`setenv GGD_VERSION 1.9.1`". Obsolete calls to *imasdb* have been removed. If compiling with the older ITM CPO infrastructure, the relevant compiler flag is `ITM_ENVIRONMENT_LOADED` which should be set by default when using the ITM WPCD framework. New IMAS features include the write-up of the *radiation*, *numerics* and *summary* IDSs, more detailed information about the species used in the simulations, and additional traceability data, most of which was introduced with version 3.22 of the IMAS Data Dictionary. All topologies supported by SOLPS-ITER can now be written in IDS format, and these are populated to the fullest extent practicable, including on all grid subsets of proper dimensionality. The geometry identifiers in **Carre** have also been adjusted to match the ones from the GGD definitions in IMAS. Throughout the SOLPS-ITER utilities, and following the new IMAS general policy, we have phased in the *database* argument name, to replace the mixture of *device*, *machine* or *tokamak*.

It is now also possible to supersede, or supplement, the IDS identification information (shot number, run number, user name, device name, and IMAS version) that are normally found in the *b2mn.dat* file by means of command line options, using a syntax like: `b2run –o "-s 123456 –r 1" b2_ual_write`.

Moreover, when converting an old SOLPS4.3 case by means of the "*convert_from_solps4.3*" script, the run output will also be written out in IDS format into the user's IMAS public database storage space.

Two new utility programs **ids2dg** and **dg2ids** allow communication between **DivGeo** and IMAS *equilibrium* and *wall* IDS descriptions. **ids2dg** translates an *equilibrium* IDS and/or a *wall* IDS description into a **.equ* and/or a **.ogr* file, respectively, containing the equilibrium and/or template information in the format needed by **DivGeo** to import. Conversely, **dg2ids** writes the content of a **.ogr* template file into a *wall* IDS description. One should build the wanted template file by selecting the elements to be included in the IDS, and then using the Export **DivGeo** feature. A fuller description of the options and arguments for these commands is available as part of the Doxygen documentation file: *$SOLPSTOP/modules/DivGeo/equtrn/doxygen/refman.pdf*.

A fuller presentation of the new IMAS SOLPS-ITER capabilities (as of early December 2019) can be found at: [https://user.iter.org/?uid=29Z3UT](https://user.iter.org/?uid=29Z3UT) .

### g. Chords for line-integrated signals

If chords are defined in **DivGeo**, they will now be included in the *input.eir* file, block 12. If, in addition, the chords are defined to be part of a `Line integrals` group, this group will be written out, as `<Set Identifier>.chr`, which is a chords file usable by **b2plot** for plotting and post-processing. Because the specifying data in **DivGeo** is not sufficient to specify all the **Eirene** options, the block 12 still needs to be completed further by the user.

### h. Eirene reaction extrapolations:

Eirene now computes extrapolations for the reactions listed in the input file for both the temperature and density ranges. Parameters for these extrapolations can be provided in the input file. This means two additional (optional) real numbers at the end of each reaction card line in block 4 of the Eirene input file. See the Eirene manual for details. A few additional reactions have also been included in the AMJUEL database.

Eirene now enforces a minimal density for the reaction rates of $10^8$ $cm^{-3}$. The default lower limit of densities in **b2ar** has been adjusted to the same value and the number of default density values in the *b2frates* tables have been increased from 16 to 32 and now range from $10^{14}$ to $10^{22}$ $m^{-3}$. The default settings for the plots in **b2yr** have been adjusted accordingly as well.

Moreover, it is now possible to provide the reaction data in so-called "Arrhenius format", for which temperature extrapolations are no longer needed. This format is recognizable in the Eirene A&M data files by having a "`−1`" coefficient by itself on the first line of fitting parameters for the given reaction. The plots of the Eirene reactions rates in both **b2plot** and **amds** utilities now also include the automated extrapolations that would be applied if no superseding instructions are found in the Eirene input file.

Some common elastic collisions are now also provided (in the *AMMONX* data file).

**i. <u>New run comparison script:</u>**

A new executable *check_b2_output.exe* is now provided in the default B2.5 compilation which can be used to compare the output of two B2.5 runs. This program is called upon by the *run_checks.sh* and *run_omp_test.sh* scripts. The user should consult the latter script for an example of the syntax to use.

**j. <u>New reference cases:</u>**

New cases have been added to the available examples in the *runs/examples* directory. To obtain the full listing, consult the manual, or, in the *runs/examples* directory, run "*make help*". To obtain any of these cases, use the "*make [case_name]*" command.

**k. <u>New environment configuration files and job submission scripts:</u>**

Many *setup.csh.*.**, *setup.ksh.*.** and *config.*.** files have been updated to accompany the various changes described in this document. In particular, the ITER, JET, ITM and IPP files have been changed to adapt to the new cluster environments at these locations, including supporting new compilers when available. Similarly, the job submission scripts have been rewritten to access the correct queues and use the SLURM framework instead of QSUB. New submission options allow for setting the job's CPU time limit, using the "*-T*" option, and requested memory size (and/or the queue to use, in some cases), using the "*-M*" option. A "*-h*" option allows for a listing of allowed syntax. At ITER, the NAG Fortran compiler is now supported, when setting the COMPILER environment variable to "*nag_f90*". The Cray compiler is also now supported, by setting COMPILER to *"cray"*. New configuration, submission and setup files were also created for the MARCONI machine when not in the WPCD ITM environment (HOST_NAME = "*MARCONI*"), as well as for the CUMULUS cluster at CCFE, the CORI cluster at NERSC, the IRIS cluster at GA, the JFRS cluster in Japan (which also has its own submission script *jfrssubmit* and template submission files *SLURM.jfrs**), and for DLUT. The IPPITM environment files have been removed as this denomination is no longer used and superseded by IPP. If specifying a custom *HOST_NAME* value by means of the *$SOLPSTOP/SETUP/setup.csh.HOST_NAME.local* file, that file must contain the single line: `setenv HOST_NAME "your_host_name"`.

A new Makefile dependency has also been added for all code executables on the *setup.csh* and *setup.csh.$HOST_NAME.$COMPILER* files, so that, when these are changed, for instance because of loading an updated module version, all the SOLPS-ITER executables are re-linked. This does not cause a full recompilation, but rather only a recall of the linker commands.

It is now possible to ask for multi-threading of the compilation using the *"-jN" make* option, where *N* is the desired number of threads. However, because the compilation of SOLPS-ITER as a whole is complex and requires a certain order in its steps, some of which must remain serial, the correct syntax to pass on such an option is: "*gmake MAKE_OPTIONS=-j8 solps"* (from the top SOLPS-ITER directory). Some of the longest routines to compile, but that are only invoked once per run and therefore do not offer an optimization advantage, namely *b2ytdr*, *b2plot*, and *b2mod_ual_io* now have embedded *NOOPTIMIZE* compiler directives to speed up compilation some more. These are activated with the *"-DNO_OPT"* pre-processor definition. The compilation has also been streamlined to automatically detect if the NCAR, GKS, and Motif libraries are available and adapt the list of targets accordingly.

The "*b2run*" script now supports a "*-o*" option that can be used to pass on further run options to the run Makefiles, when these are surrounded in quotes. Additionally, if wanting to launch a run with the default debugger (defined by the *DBX* environment variable), it is no longer necessary to specify the debugger's name after the "*-d*" option.

### l. New case conversion features:

The **b2yt** conversion program will now ensure that the BGK species listed as part of the background species in the **Eirene** input file are at the end of the list and re-order the background species and associated variables as necessary to achieve this. When removing species, the BGK species and associated reaction lists will also be adjusted. The special case when all hydrogenic species are removed is now also treated properly.

When converting a SOLPS4.3 case that contains He, N or Ar impurities, the **b2sxdr** program now provides default values for the electron- and ion-impact collision cross-sections used in *b2ah.dat* to compute transport coefficients for neutrals. The **b2ah** program also now checks that numbers for these cross-sections are given, to avoid infinitely large transport coefficients for these neutral species.

### m. Miscellaneous new switches:

Several new switches have been introduced, among which are:

- "*my_out_digits*" specifies the number of significant digits printed out in the *output/*.dat* files. Defaults to 6 in normal mode and to 15 in debug mode.

- "*b2mwti_jsep*" can be used, for slab geometry cases only, to indicate the position of the separatrix for output plotting purposes. For all other topologies, the position of the separatrix is, as before, computed automatically by the code from the connectivity arrays.
- "*b2trcl_csig_mltpl*" can be used to multiply the electric conductivity between core guard cells to damp out large currents that may arise from a poloidal variation of the electric potential along the core boundary.
- "*get_residuals*" can be used to obtain the residuals of all equations, even if they are not solved, when set to "*1*". It is also possible now to turn off or on the total parallel momentum and energy equations by means of the *SOLVEMT* and *SOLVEET* array of switches, included in the *b2.numerics.parameters* file, in the same way as for the other equations such as *SOLVEMO*, etc… The total equations will now only be solved if all underlying components are also solved.
- The CFL pressure gradient limiting, which is applied to minority species and fluid neutrals, can now be turned off if one sets "*b2sigp_pressure_restriction*" to "*1*".
- *"b2mndr_zn_rescale"* can be used to specify the nuclear charge of the isonuclear sequence of species which density is to be rescaled by *abs(density_rescale)* if *"b2mndr_density_rescale"* is set to a negative value. In older code versions, this was only possible for Helium.

In the Eirene input file, there is a new TRCHKTIM option to trace how the spent CPU time is split per particle type, species, and stratum. This switch is located in the first line of block 11 of the input file.

### n. <u>Other new scripts:</u>

Several new scripts have been added to the distribution, such as "*awk.reverse_columns*", "*SLURM.postprocess*", "*SLURM.postprocess_coupled*", "*plot_rzpsi.dat*", "*which_latex*", "*which_pdftops*" and, for monitoring external sources, "*b2sext_s*[*ch*/*he*/*hi*/*mo*/*na*]*_reg*[*_region*]" and "*fchxreg*" and "*fchyreg*" for the currents. There are also a few new Python scripts, such as "*currents.py*", "*fluid_core_fluxes.py*", and "*particle_balance.py*". "*calculate_sputter_inner_target.py*" and "*calculate_sputter_outer_target.py*" allow for a post-processing calculation of sputtering fluxes for a given plasma background incident on a target.

SOLPS-ITER now expects to have Python3 available and all scripts have been converted for compatibility. New scripts provided by users will need to be follow that rule as well. The aliases "*solps*", "*solps_doc*", "*solps_help*", "*eirene*" and "*b2*" have been introduced (at least on ITER systems) for linking to SOLPS-GUI functions.

The script “*inspect_plasmastate*” now allows the user to find out at what plasma time a given *plasmastate.XXXX* save-file was written, to help in choosing which file they want to use when rebuilding an interrupted or crashed case.

5. **Changes to the output:**

   a. **Move of XMCT timing array from *fort.11* to *fort.14*:**

The Eirene array `XMCT` which contains timing information for the strata used during MPI parallelization has been moved from the *fort.11* to the *fort.14* file. Accordingly, **Uinp** also prepares the `NFILE` switch so that MPI runs will create that file, **b2mndr** also makes sure checkpoint versions of that file are created, and the *NFILE* conversion rule in **b2ytdr** has been adjusted.

   b. **New tracing output files:**

There is a new tracing output file, *tracing/blnn_SPb.trc*, which contains information related to particle balance terms relevant for the new code speed-up schemes. This file is treated in the same way as the other tracing files and written by the *b2blns* routine. A description of the contents of that file is available in *doc/b2blnc_SPb_description.pdf*.

A description of the Matlab balance analysis routines provided by David Moulton of CCFE (which have themselves been updated to include terms from drifts) has been added to the documentation as *doc/Balance_routines_in_SOLPS-ITER.pdf*.

Many of the optional output files have been renamed (and several more added) to give them more telling names, and they are now listed in the SOLPS-ITER manual, Appendix G. Many of the individual terms from the model equations can be written out in these files, and the description in the manual has been updated to link the terms in the equations to the output files containing them. Thanks to Ilya Senichenkov and Lisa Kaveeva in St. Petersburg for doing the heavy lifting for this feature. A new discussion on energy conservation has also been included in the manual.

   c. **Eirene spectral lines:**

The input block 12 in the Eirene input file has been redone so spectral lines can be defined with multiple contributions from several species, and line-of-sight diagnostics have been updated. See the Eirene manual for details. Conversion rules for these spectral line definitions in **b2yt** have also been introduced. To obtain the emissivity for the default hydrogenic lines, the recommended method is now to add the two lines:

```
DEFINE_LINES
     0     1
```

at the beginning of block 12 of the Eirene input file. This is now the default **Uinp** behaviour. New collisional-radiative models have been introduced as well: “`CONSTANT`”, “`MULTIPLY`” and others now require additional data (please consult the Eirene manual).

The routines *ba_alpha*, *ba_beta*, *ba_gamma*, *ba_delta*, *ly_alpha* and *ly_beta* have been revisited as well.

This new treatment deprecates the “*eirene_lhalpha*” and “*eirene_lvib*” input switches in *b2mn.dat* and these must now be removed from that file. They are removed when running **b2yt** to convert an older *b2mn.dat* file.

The photon reflection model has also been changed and the *graphite_ext.dat* and *mo_ext.dat* files are no longer needed by the code. They have been removed from job submission scripts and the default input file build-up. The `LHABER` and `NEXVS` flags are also deprecated.

### d. Eirene radiation tallies:

The Eirene output now includes, in the *fort.44* file, two new arrays, namely *emolrad* and *eionrad*, corresponding to the radiation due to the molecules and molecular ion species followed by Eirene, separated per species. In addition, the *eneutrad* array now is also separated by atomic species. The format identifier for files including these arrays is `20180323`. Conversion rules in **b2yt** have been added for these arrays. These tallies are numbered are volumetric tallies 101 to 103 in Eirene (radiation from atoms, molecules, and molecular ions, respectively).

### e. Eirene standard output during MPI runs:

When running Eirene with MPI, the output from each slave node is redirected to an *output.XXXX* file. By default, only output from the last Eirene call will be present in such files. It is however now possible, by means of the *LOUTAPP* variable, defined in *modules/Eirene/src/user-routines/user_iter/defaults_usr.F*, to append the output from successive Eirene calls in these files, if the variable is set to `.true.`.

### f. Other additional b2plot output:

The total heat flux provided by **b2plot** and other diagnostic scripts now contains an additional factor of *fna*Ti*. Various output routines and files have been adapted to limiter configurations and/or slab geometries. For these options to work best, it is important to have properly provided the *LTNS* array in *b2.neutrals.parameters* to indicate the location of the recycling targets in computational space.

Other new **b2plot** quantities and features include:

- The number of Eirene species “*natm*”, “*nmol*” and “*nion*”.

- The poloidal and radial electron fluxes "*fnex*" and "*fney*".
- The plasma pressure "*pz*", summed over the charged species only.
- The Mach number "*mach*" and dynamic pressure "*p**" calculations no longer include contributions from neutral species to either the pressure or the mass density.
- The toroidal and poloidal flux functions ("*ffbz*" and "*psi*", respectively) can now be plotted, provided they have been written out by **b2ag** in the *b2fgmtry* file. In SOLPS-ITER v.3.0.7, the *traduit* program from **Carre** now writes out this information as part of the standard run set-up workflow. It is then also possible to plot radial profiles against the poloidal flux $\psi$ by using the "*fpsi*", "*mpsi*" and "*rpsi*" **b2plot** functions, which, respectively, plot against the full $\psi$ value, the normalized $\psi$ value, and the square root of the normalized $\psi$ value.
- The constant and variable parts of the source arrays "*sna*", "*sne*", "*smo*", "*smq*", "*she*", "*shi*", "*sch*", "*smaf*", "*smag*", and "*smav*" can now be extracted by providing an integer (between 0 and 3, corresponding to the component index) before the variable name to be plotted. If no integer is provided, then the total source will be given.
- A new "*nice*" command (`on` by default) toggles the setting of the axes limits of plots to round numbers, so as to allow for better control of the appearance of the **b2plot** graphs.
- It is now possible to obtain thicker graph frames, labels, axes and ticks by means of the "*bold*" command (which toggles this behaviour on and off) and to change their colour with the "*fcol*" command.
- The new "*glsz*" command allows for changing the size of the global header that can be added at the top of the plot pages. The "*lbsz*" label size command applies only to the labels and titles of individual graphs.
- The "*neutrad*" array is now dimensioned with the number of Eirene atomic species. Radiation due to Eirene molecule and molecular ion species are now available as the "*molrad*" and "*ionrad*" arrays. Their contributions are also added to the total radiation calculations used elsewhere in **b2plot**, including for the power load to the targets.
- A new "*etal*" function allows for plotting of data obtained from 2-D tally output in block 11 of the Eirene input file, by providing the name of the output file as a string argument. The data is expected to be provided on the Eirene triangular grid and will be plotted on that grid.
- It is also now possible to use the NCAR math symbol font in labels with the *'\m'* prequalifier (terminated with *'\d'* to return to the default font). For example, *'\mU\d'* will yield an *'×'* cross-product symbol.
- The command *"eirc"* copies Eirene data into the B2.5 fluid arrays. A safety has been added so that the copy will only be performed once. In addition to

the neutral density itself, the mass density array *"ro"* is also updated. If the flow field arrays *"pfla"* and *"rfla"* are available, then the fluid fluxes *"fnax"* and *"fnay"* and the velocity fields *"ua"*, *"up"* and *"vv"* are updated to include the Eirene neutral flows. The associated default settings for *"uutraj"*, *"vvtraj"*, *"uuse"* and *"vvse"* follow the same changes.

When asking for divertor neutral pressure with the *"# user"* command in **b2plot** (in which case # must be an integer between *ny* and 1000, exclusively), the output, available in the *b2pl.exe.dir/pressure.ntr* file, will contain, on the first data line, the total pressure due to all neutral species, and then each succeeding line will provide the partial pressure corresponding to each individual neutral species contributing to the total. Of the three columns, the first is the average divertor pressure, the second is the pressure on the inner side of the divertor, and the third is the pressure on the outer side. The average is weighted by the surface area of contact between the HFS and LFS of the PFR boundary regions used for computing the average, as defined in *b2.user.parameters*. For the bolometer output, ("*-13 user*" option, producing the "*radsources.dat*" file), the location electron temperature, ion temperature, electron density and species densities have been added. The pressure obtained with the "*-8 user*" command is now only the plasma pressure, not including contributions from neutral species.

In the "*peak_power*" file (obtain by means of *"# wlld"* where *#* is an integer larger than or equal to 100), the values of the peak electron temperature and density along the targets have been added to the contents. The power carried back by recycled neutrals as they leave the targets is now removed from the code output when computing the target power fluxes. For the target load output files, the split of the plasma load between thermal and recombination contributions is now provided.

The Summers database files have been added to the main distribution. This recovers the use of the "*spec*" command for obtaining selected line radiation intensities. The code has also been modified to be compatible with the new internal format of ADAS files.

### g. New *vertex_data.out* connectivity file:

In order to speed up the start of most B2.5 programs, the geometry build-up now includes the creation by **b2ag** (or any subsequent B2.5 program if the file is not yet present) of a *vertex_data.out* file that contains connectivity information of the grid, instead of having it recomputed every time.

### h. New output tallies and case archiving features:

The radiation emitted by the neutral species followed by Eirene is now available among the code tallies as "*rdneureg*", including in the NetCDF output.

The batch-averaged data, using with the Eirene averaging speed-up scheme, are now written out in a separate *b2batch.nc* file, and no longer included in *b2time.nc*. All relevant file manipulation scripts for preparing, recovering, or post-processing runs have been modified to include this new file when present.

The version number of the MDsplus data files being saved has been increased to 28 to reflect the changes in output data definitions described elsewhere in this document.

When saving run output into the MDSplus databse, the definition of *dspar* (parallel length of a cell) has been corrected. Later, the *emolrad* and *eionrad* arrays, containing the radiation due to Eirene molecules and molecular ions, respectively, have been also added, bumping up the *dbversion* number to 27. Version 28 corresponds to the usage of the corrected total heat fluxes from *b2xpen*.

When using the *save_mds* and *resave_mds* scripts to write to the MDSplus database, it is now possible to pass along some variable redefinitions to the underlying *Makefile.plot* call with the syntax: `-r VARIABLE=NEW_VALUE`.

### i. **Basic *nc2text* utility and NetCDF support:**

If the local environment does not provide the *nc2text* utility program to read NetCDF files (but a NetCDF library is available), the standard code distribution now provides a basic alternative program (*nc2text_simple*). If the Makefile detects that the utility is not available, that alternative program will be compiled and made accessible to the code scripts needing to read the NetCDF output files in a user-transparent fashion. (Thanks to Jeremy Lore from ORNL for providing this functionality).

SOLPS-ITER now requires NetCDF version 4 or higher to compile. This change was made necessary to avoid problems when multiple code runs on the same machine where writing NetCDF files simultaneously. It is now also possible, by means of the *"b2mndr_cdf_default"* switch, to change the internal formatting of the NetCDF files from the local library default. This also means that, if you have an *nc2text* utility inherited from SOLPS5.0, which was built with NetCDF3, you need to remove it and let the Makefile rebuild it.

## 6. **Getting a case restarted with v. 3.0.7:**

As a consequence of all the changes above, in order for users to take full advantage of the new 3.0.7 features, users wishing to continue their older runs should proceed in the fashion detailed below (and in the same order). Of course, if converting older cases or building new cases from scratch, the **Uinp**, **b2yt** and **b2sxdr** programs have been modified to produce input files already compatible with v. 3.0.7, which should not preclude a visual inspection of all relevant input files, as is best practice.

1. Re-run **Carre** to add the magnetic field information to the grid files. Then re-run *triang* to make sure the triangles are still matching the Carre grid.
2. Re-run **b2ag** (in your baserun/ directory, after checking that you have a valid *b2ag.dat* file and deleting *b2ag.prt*) to obtain a new *b2fgmtry* file where the magnetic contact area and face angles have been recalculated.
3. Re-run **b2ah** (in your *baserun/* directory)
4. Re-run **b2ai** (in your *baserun/* directory), to update your *b2fstati* file. Because you have done this, remember to do "*touch b2fstati*" **afterwards** in **each** of the case directories on the parallel level to your *baserun/* directory!!!
5. Remove deprecated switches, such as *"eirene_lhalpha"* from your *b2mn.dat* file. If using drifts, consider updating *"b2tfnb_drift_style"* to *"2"*.
6. Check your Eirene input files to make sure that the value of `NFILE` matches your intended use of *fort.1[0-5]* files, in particular that you read and write *fort.13* if using BGK species, *fort.14* if using the `BALANCED` MPI parallelization scheme, and *fort.15* if you use a time stratum. Refer to the Eirene manual if you are unsure of which setting to use. Consider using the `NPRLL` switch to specify your chosen MPI parallelization strategy and modifying block 12 to obtain hydrogen line emissivities.
7. If you are restarting a case that was computed using code from the *master* or *develop* branches, or the *release/3.0.7* branch from before November 2019, you should delete the existing *fort.1[0-5]* as there is an internal change in Eirene that has modified the content of these files.
8. If you were previously using NetCDF3, you may not be able to append to your older **.nc* files and would then need to delete/rename them.

## 7. Additional notes for developers:

i. Two new solvers (based on 7-point and 11-point stencils, respectively) have been added for the computation of terms related to the perpendicular viscosity current. Additionally, the Pfirsch-Schlüter and perpendicular viscosity terms are now computed in standalone routines *b2tepsch*, *b2tipsch*, *b2tnpsch*, *b2txnpsch*, and *b2txvsprco*.

ii. Several Eirene routines have been merged or consolidated into a single file, such as (*geomd*, *geomd_linda*, *indmap*, *indmpi*, *neutr*) into *couple_specific.f*, (*updatm*, *updmol*, *updion*) into *update.f*, (*fpatha*, *fpathm*, *fpathi*) into *fpath.f*, (*algebr*, *fehler*, *klamme*, *mecker*, *operat*, *oprand*, *rdcn*, *ruksub*, *schrit*, *signok*, *subtit*, *zerleg*) into *eirmod_algebra.f*.

iii. The Eirene routines *ba_alpha*, *ba_beta*, *ba_gamma*, *ba_delta*, *ly_alpha* and *ly_beta* are now obsolete and these line emissivities are now computed by the *setup_default_emissivity* routine. The modules *eirmod_caprmc* and *eirmod_ccflux* and the *scatang* routine have also been removed as the photon transport is now done differently.

iv. A new integer array *micmp(i,j)* which contains the number of atoms *i* in Eirene molecular ion *j* is now available for use. It is computed internally after either the first read of the *fort.44* file or the first call to Eirene.

v. As part of the compatibility work for NAG and Cray compilers support, a lot of code has been rewritten to avoid obsolescent Fortran features, such as DO loops not ending in ENDDOs, using KIND declarations throughout instead of REAL*4 or REAL*8, replacing computed GOTOs by SELECT CASEs and arithmetic IFs by normal IFs, defining more variables explicitly, removal of 'H' format descriptors, avoiding type conversions in intrinsic operators, etc… This means a new "*CARRETYPES.F*" include file to provide these definitions. Code specific to the NAG (Cray) compiler is bracketed by "*#ifdef NAGFOR*" ("*#ifdef CRAY*") compiler commands. KINDs I4 and I8 were introduced when needing to specific short or long integers explicitly. It is asked that future development continue along the same lines. A new R16 kind was also introduced for quadruple-precision floating-point operations, required when solving very large matrices. If quadrupole precision is not available on your system, the *–DNOQUAD* compiler option can be used to revert back to double precision algebra.

vi. The version of the MSCL library distributed by ITER has been revised to simplify its installation. Please keep in mind that it is only known to work if compiled in single-precision, and that real arguments passed to MSCL routines MUST be single-precision as well.

vii. The Makefiles have been modified so that, when building up the dependencies lists, it is now possible inside the source code to use the "*only*" qualifier when using a module on the same line as the "*use module*" commands.

viii. A new environment variable *FFLAGSEXTRA*, which can be set at the level of the *$SOLPSTOP/SETUP/config.$HOST_NAME.$COMPILER* files, is available to pass on system-wide compilation flags to all FORTRAN executables of the SOLPS-ITER suite.

ix. The *b2plot/page.F* routine has been renamed as *b2plot/b2page.F* to avoid a name clash with the Eirene routine *assistant/page.F*.

x. The *streql* routine has been modified so that blank characters are considered equivalent to spaces.

xi. A new routine *b2xprz* has been added to compute the ion mass density *rz*, where only contributions from charged species are included.

xii. When compiling in debug mode with GFortran, the flags *"-ffpe-trap=invalid -ffpe-trap=zero –ftrapv –fsignaling-nans*" have been added.

xiii. To allow for system flexibility, Fortran default file names are now coded as "*fort*//[unit number]" (for uppercase names) or "*fort_lc*//[unit number]" (for lower case numbers) and defined in *eirmod_cinit.F* (for Eirene) and *b2mod_eirbra* (for B2.5).

xiv. The random number generator in Eirene has been updated. The associated routines have been extensively rewritten.

xv. The *XMCT* timing array has been moved from the *eirmod_coutau* to the *eirmod_cai* modules. The *NRPES* variable is deprecated and replaced throughout by *NPRS*.

xvi. Eirene now makes use of "*ERROR STOP*s" in *exit_own.f* to allow for easier management of CI test failures. These are not supported by all compilers, so preprocessing pragmas are included to prevent compilation errors.

xvii. To avoid compilation issues with the **Uinp** block data file *uiblkd.F* when importing external module data, a new *eirmod_solps.F* module has been created that contains only the parameters needed to be passed and no extraneous variables.

xviii. The FORTRAN code in Eirene has been modernized, whereby use of computed IF and GOTO statements is avoided as much as possible, routine containing entries have been converted to modules, and alternate returns have been replaced by output arguments.

xix. On some implementations, the serial compiler and the MPI compiler may not be based on the same architecture. For instance the serial compiler is the Intel-based *ifort*, while the MPI compiler is the *gfortran*-based *mpif90*. For such cases, the configuration files have been adapted to detect which compiler architecture is being used and adjust the compilation flags and library linking instructions accordingly.